Atakan Eksi

On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam, the work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

1-) Consider the matrix $A=\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$ some of the elimination matrices are given below

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} , \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$$

a) $\alpha=?\ \beta=?\ m=?$

From the second eliminations matrix, we have

$3\alpha - 8 = 0 \rightarrow \alpha = 8/3$

$2 + 2m = 0 \rightarrow m = -1$

From the first elimiations matrix given, we have

$\beta - 8/3 - 1 = 0 \rightarrow \beta = 11/3$

$34/3 + k = 0 \rightarrow k = -34/3$

Since a matrix after elimination becomes

$$\begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & 1 & 3\gamma/49 & 6/49 \\ 0 & \beta - 8/3 & 3+\gamma & 40/9 \\ 0 & 34/3 & -6+\gamma & -2/3 \end{bmatrix}$$

Then after elimation using third matrix given we have

$$A \begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & 1 & 3\gamma/49 & 6/49 \\ 0 & 0 & 3+\frac{46\gamma}{49} & \frac{40}{9}-\frac{6}{49} \\ 0 & 0 & & \end{bmatrix} \qquad \text{so} \quad \frac{46}{49}\gamma = -2$$

$$\gamma = \frac{-98}{46}$$

1.b) then becomes

$$\begin{bmatrix} 1 & 8/3 & 98/46 & 8/3 \\ 0 & 1 & -6/46 & 6/49 \\ 0 & 0 & 1 & \frac{40}{9}-\frac{6}{49} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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2-) A geometry in 'k' dimensional space is given by $x = 5y$ expression, where 'x' and y represents the axis. The intersection with another geometry gives a volume

a. Give the minimum dimension of the space with your valuable comments?

(a) Geometry is a k-1 dimensional shape Thinking the problem in we get a plane from the intersection of a plane and volume. So the geometry could be either plane or volume.

$$k - 1 = 2$$
$$k = 3$$

(b) We can use eq. of a plane in 3D to represent volume in 4D space

$$x + y + z = 1$$

Unknown geometry as explained above is a volume

(c)
$$\begin{array}{r} x + y + z = 6 \\ + \; x + 4y - z = 1 \\ \hline 2x + 5y = 7 \end{array}$$

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3a) $H^2 = (I - 2uu^T)^2 = I^2 - 4uu^T + 4uuu^T u u^T$

Since $u^T u = 1$ we have

$H^2 = I^2 - 4uu^T + 4uu^T = I^2 = I$

b) $\det(H) = \det(I_n - 2uu^T) = I - 2u^T u$

knowing that $u^T u = \|u\|^2 = 1$, we have

$\det(H) = 1 - 2 = -1$

So H is orthonormal matrix

c) $Hu = \lambda u$

$Hu = (I - 2uu^T)u = u - 2uu^T u = u - 2u = -u$

$-u = \lambda u$

$\lambda = -1$

d) Pick a vector orthogonal to u, say v, we have

$Hv = (I - uu^T)v = v - 2uu^T v = v - 2u \cdot 0 = v$

Then associated eigenvalue is 1. Space spanned by u has a orthogonal complement of size $(n-1)$. So there is a basic of orthogonal eigenvectors with the same eigenvalue which is 1. If we add u, to have n vectors we can conclude H is diagonalizable

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4) Two tanks are connected, first tank has $v(0)=10$ liters liquid...

a) change in tank 1 = − change in tank 2

$\frac{dv}{dt} = -(v-2w)$ ← proportional to this term

$\frac{dw}{dt} = (v-2w)$

$$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$$

b) Let $\begin{bmatrix} v \\ w \end{bmatrix} = x'(t) = A\,x(t)$

$x(t) = e^{At}\,x(0)$

$e^{At} = (I + At + \frac{A^2t^2}{2!} + \dots\dots)$

eigenvalues of A are λ; $\lambda^2 + 2\lambda = 0 \Rightarrow \lambda = 0$ and $\lambda = -2$

corresponding eigenvalues are $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ for $\lambda = -2$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ for $\lambda = 0$

Then $e^{At} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} e^{0t} & \\ & e^{-2t} \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}^{-1} =$

$= \begin{bmatrix} 1 & -e^{-2t} \\ 1 & e^{-2t} \end{bmatrix}\begin{bmatrix} 0,5 & 0,5 \\ -0,5 & 0,5 \end{bmatrix} = \begin{bmatrix} 0,5 + 0,5e^{-2t} & 0,5 - 0,5e^{-2t} \\ 0,5 - 0,5e^{-2t} & 0,5 + 0,5e^{-2t} \end{bmatrix}$

$x(1) = e^{A}\,x(0)$

$= e^{A}\begin{bmatrix} 10 \\ 40 \end{bmatrix} = \begin{bmatrix} 25 - 15e^{-2} \\ 25 + 15e^{-2} \end{bmatrix}$

c) A steady state, $t \to \infty$ is we would expon. $\to 0$

So, $e^{A\infty} = \begin{bmatrix} 0,5 & 0,5 \\ 0,5 & 0,5 \end{bmatrix}$, $v(\infty) = 25$, $w(\infty) = 25$

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